

Practical Assignment: 3

Problem Statement

A university wants to automate the process of managing student academic and administrative information. The system should accept student details, subject marks, attendance percentage, and fee payment information. Based on the provided information, the application should calculate results, determine examination eligibility, and generate a detailed performance report.

The system should be developed as a console-based Java application using appropriate data types, variables, operators, decision-making statements, loops, arrays, and command-line arguments wherever applicable.

Module 1: Student Information

The system should collect the following information:

Enrollment Number
Student Name
Program Name
Semester
Academic Year

Module 2: Subject Marks Management

The university follows a five-subject evaluation system.

The application should:

Store marks of five subjects using an array.

Validate that marks are between 0 and 100.

Calculate:

Total Marks

Average Marks

Percentage

Module 3: Grade Evaluation

Assign grades according to the following criteria:

Percentage	Grade
90 and Above	A+
80 – 89	A
70 – 79	B
60 – 69	C
50 – 59	D
Below 50	F

The application should display the final grade.

Module 4: Attendance Eligibility

The university requires a minimum attendance of 80%.

The application should:

Accept attendance percentage.

Determine whether the student is eligible to appear in the examination.

Conditions

If attendance ≥ 80 Display eligible for Examination Otherwise Not Eligible for Examination

Module 5: Fee Payment Status

The annual tuition fee is ₹75,000.

The application should:

Accept the amount paid by the student.

Calculate the remaining fee balance.

Display: Fully Paid, Partially Paid, Pending based on the fee payment status.

Module 6: Academic Performance Analysis

The application should identify:

Highest Subject Marks

Lowest Subject Marks

Number of Subjects Passed

Number of Subjects Failed

Passing Marks = 40

Use loops and arrays to perform the analysis.

Module 7: Scholarship Eligibility

A student becomes eligible for a scholarship if:

Percentage ≥ 85

Attendance ≥ 90

No Subject Failed

Display: Scholarship Eligible or Scholarship Not Eligible

Module 8: Report Generation

Generate a formatted report showing:

STUDENT PERFORMANCE REPORT

Enrollment No :

Name :

Program :

Semester :

Marks Obtained :

Subject 1 :

Subject 2 :

Subject 3 :

Subject 4 :

Subject 5 :

Total :

Average :

Percentage :

Grade :

Highest Marks :

Lowest Marks :

Attendance :

Exam Eligibility :

Fee Paid :

Balance Fee :

Payment Status :

Scholarship Status :

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Problem Statement

A retail store wants to develop a Java-based Sales and Inventory Analysis System to record products sold during the day and generate a daily sales report.

The application should accept product information, calculate individual and total sales, apply discounts, identify the best-selling product, analyze inventory levels, and generate a final sales summary.

The application should use arrays, loops, operators, decision-making statements, variables, data types, and command-line arguments.

Module 1: Store Information

The store name should be accepted as a command-line argument.

Example: java RetailStore "SmartMart"

The program should display an appropriate message if the store name is not supplied.

Module 2: Product Information

The store maintains five products.

For every product, store:

Product ID

Product Name

Price

Quantity Sold

Available Stock

Use suitable arrays.

For example:

productId[]

productName[]

price[]

quantitySold[]

stock[]

Module 3. Product Sales Calculation

For each product:

Product Sales = Price × Quantity Sold

Calculate the total sales of all products.

Module 4. Discount Calculation

Apply discount based on total product sales:

Sales Amount	Discount
Below ₹1,000	0%
₹1,000–₹4,999	5%
₹5,000–₹9,999	10%
₹10,000 and above	15%

Use decision-making statements.

Module 5. Final Sales Amount

Calculate:

Discount Amount = Total Sales × Discount Rate

Final Amount = Total Sales - Discount Amount

Module 6. Inventory Analysis

After sales, calculate:

Remaining Stock = Available Stock - Quantity Sold

Identify:

Products with low stock

Products that are out of stock

Products with sufficient stock

A product should be considered Low Stock if remaining stock is less than 10 units.

Module 7. Best-Selling Product

Using an array and loop, identify the product having the highest quantity sold.

Display:

Best-Selling Product :

Quantity Sold :

Module 8. Sales Performance

Determine the store's sales category:

Total Sales	Category
Below ₹5,000	Low
₹5,000–₹9,999	Moderate
₹10,000–₹19,999	Good
₹20,000 or above	Excellent

Module 9. Daily Sales Report

Generate a report such as:

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SMARTMART SALES REPORT

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Product ID	Product	Price	Qty	Sales
101	Laptop	50000	2	100000
102	Mouse	500	5	2500
103	Keyboard	1200	3	3600

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Total Sales :

Discount :

Discount Amount :

Final Sales Amount :

Best-Selling Product :

Quantity Sold :

Low Stock Products :

Out of Stock :

Sales Category :

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